

Develop embedded software for the thermostat control

HW:

One ADC input for temperature measurement
0-5V with linear relation: 0V - 0C, 5V - 50C

One output Dout for heater control
Dout = 0 - heater on
Dout = 1 - heater off

Thermostat

The thermostat should use PID control.

Settings of the thermostat

Settings can be changed using text terminal over UART (USB)

List of settings:

T set - Temperature to be achieved and maintained

Thermostat - thermostats status - ON/OFF. If switched off, the heater is not controlled and switched off, only temperature monitored.

PID coefficients Kp, Ki, Kp

Monitoring of the thermostat

Over the serial communication and using terminal, information about thermostat should be read at any time:

T - > current temperature in C

Tstat -> thermostat status - on or off

T set - set temperature

PID coefficients Kp, Ki, Kp

Expected result of practical task

Source code for freely selected MCU STM32, source code should be written in C, code documented and submitted to the provided GIT repository.

Optional, if you have a suitable Dev board - working software on the dev board represented during the next meeting.